
Question No: 1 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) > 0$ then f has relative At x_0

- ▶ Minima
- ▶ Maxima
- ▶ None of these

Question No: 2 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) < 0$ then f has relative At x_0

- ▶ Minima
- ▶ Maxima
- ▶ None of these

Question No: 3 (Marks: 1) - Please choose one

A line $y = y_0$ is called a for the graph f if $\lim_{x \rightarrow +\infty} f(x) = y_0$ or $\lim_{x \rightarrow -\infty} f(x) = y_0$

- ▶ Vertical asymptotes
- ▶ Horizontal asymptotes
- ▶ None of these

Question No: 4 (Marks: 1) - Please choose one

If $f(x) = 3x^8 + 2x + 1$ then $f'(x) =$ _____

- ▶ $3x^7 + 2$
- ▶ $24x^7 + 2$
- ▶ $3x^9 + 2x^2$
- ▶ $24x^9 + 2x^2$

Question No: 5 (Marks: 1) - Please choose one

$\frac{d(\tan x)}{dx} =$

- ▶ $\sec x$
- ▶ $\sec^2 x$
- ▶ $\operatorname{cosec} x$
- ▶ $\operatorname{cosec}^2 x$

Question No: 6 (Marks: 1) - Please choose one

If $xy = 4$ then $\frac{dy}{dx} =$

- ▶ 0
- ▶ $\frac{-1}{x^2}$
- ▶ $\frac{4}{x^2}$
- ▶ $\frac{-4}{x^2}$

Question No: 7 (Marks: 1) - Please choose one

If $2x - y = -3$ then $\frac{dy}{dx} =$

- ▶ 2
- ▶ -2
- ▶ 0
- ▶ -3

Question No: 8 (Marks: 1) - Please choose one

$\frac{d}{dx}[\sec x] =$ _____

- ▶ $\frac{1}{1 + \sin^2 x}$
- ▶ $\frac{-\sin x}{1 + \sin^2 x}$
- ▶ $\frac{1}{1 - \sin^2 x}$
- ▶ $\frac{\sin x}{1 - \sin^2 x}$

Question No: 9 (Marks: 1) - Please choose one

$30^0 =$ _____

▶ $\frac{\pi}{3}$

▶ $\frac{\pi}{4}$

▶ $\frac{\pi}{6}$

▶ $\frac{\pi}{2}$

Question No: 10 (Marks: 1) - Please choose one

Consider a function $h(x)$ and a constant c then

$\frac{d}{dx}((c) \{h(x)\}) = \underline{\hspace{2cm}}$

▶ 0

▶ $\frac{d}{dx}(h(x))$

▶ $\frac{d}{dx}(h(cx))$

▶ $c \frac{d}{dx}(h(x))$

Question No: 11 (Marks: 1) - Please choose one

Suppose that f and g are differentiable functions of x then

$\frac{d}{dx} \left[\frac{f}{g} \right] =$

▶ $\frac{[g][f'] - [f][g']}{g^2}$

▶ $\frac{[g'][f] - [f'][g]}{g^2}$

▶ $\frac{[g][f'] - [f][g']}{f^2}$

▶ $\frac{[g'][f] - [f'][g]}{f^2}$

Question No: 12 (Marks: 1) - Please choose one

$$\frac{d}{dx}[\operatorname{cosec} x] = \underline{\hspace{2cm}}$$

- ☐ $\frac{1}{1 + \cos^2 x}$
- ☒ $\frac{-\cos x}{1 - \cos^2 x}$
- ☐ $\frac{-\cos x}{1 - \cos^2 x}$
- ☐ $\frac{1}{1 - \cos^2 x}$

Question No: 13 (Marks: 1) - Please choose one

$$y = f(g(h(x)))$$

If

$$u = g(h(x))$$

$$v = h(x) \quad \text{then} \quad \frac{dy}{dx} = \underline{\hspace{2cm}}$$

- ☒ $\frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$
- ☐ $\frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$
- ☐ $\frac{dv}{du} \cdot \frac{du}{dv} \cdot \frac{dy}{dx}$
- ☐ $\frac{dy}{dv} \cdot \frac{dv}{du} \cdot \frac{du}{dx}$

Question No: 14 (Marks: 1) - Please choose one

Chain rule is a rule for differentiating _____ of functions.

- ☐ Product
- ☐ Sum
- ☐ Difference
- ☒ Composition

Question No: 15 (Marks: 1) - Please choose one

$$\frac{d}{dx}[x^n] = nx^{n-1}$$

The power rule, _____ holds if n is _____

- ▶ An integer
- ▶ A rational number
- ▶ An irrational number
- ▶ All of the above

Question No: 16 (Marks: 1) - Please choose one

Let a function f be defined on an interval, and let x_1 and x_2 denotes two distinct points in that interval. If $f(x_1) = f(x_2)$ for all points x_1 and x_2 then which of the following statement is correct?

- ▶ f is a decreasing function
- ▶ f is an increasing function
- ▶ f is a constant function

Question No: 17 (Marks: 1) - Please choose one

If $f''(x) < 0$ on an open interval (a,b) then which of the following statement is correct?

- ▶ f is concave up on (a, b).
- ▶ f is concave down on (a, b)
- ▶ f is linear on (a, b).

Question No: 18 (Marks: 1) - Please choose one

If $x > \frac{1}{2}$ then $\frac{d}{dx}[\ln 2x] =$ _____

- ▶ $\frac{1}{x}$
- ▶ $\frac{2}{x}$
- ▶ $\frac{1}{2x}$
- ▶ $\frac{2}{x^2}$

Question No: 19 (Marks: 1) - Please choose one

Let $y = (x^3 + 2x)^{37}$. Which of the following is correct?

- ▶ $\frac{dy}{dx} = (37)(x^3 + 2x)^{36}$

► $\frac{dy}{dx} = 111x^2(x^3 + 2x)^{36}$

► $\frac{dy}{dx} = (111x^2 + 74)(x^3 + 2x)^{36}$

► $\frac{dy}{dx} = (111x^2 + 74)(x^3 + 2x)^{38}$

►

Question No: 20 (Marks: 1) - Please choose one

If we have $x^2 + y^2 = 1$ then $\frac{dy}{dx} =$ _____

► $\frac{-x}{y}$

► $\frac{x}{y}$

► $\frac{-y}{x}$

► None of these

Question No: 21 (Marks: 1) - Please choose one

$\log_b ac =$ _____

► $\log_b a + \log_b c$

► $\log_b a - \log_b c$

► $\frac{\log_b a}{\log_b c}$

► $(\log_b a)(\log_b c)$

Question No: 22 (Marks: 1) - Please choose one

$\log_b a^r =$ _____

► $a \log_b r$

► $r \log_b a$

$$\frac{\log_b a}{\log_b r}$$

- ▶
- ▶ $\log_b a + \log_b r$

Question No: 23 (Marks: 1) - Please choose one

$$\log_b \frac{1}{c} = \underline{\hspace{2cm}}$$

- ▶ $\log_b c$
- ▶ $1 - \log_b c$
- ▶ $-\log_b c$
- ▶ $1 + \log_b c$

Question No: 24 (Marks: 1) - Please choose one

$$\log_b \frac{1}{t} = \underline{\hspace{2cm}}$$

- ▶ $\log_b t$
- ▶ $1 - \log_b t$
- ▶ $1 + \log_b t$
- ▶ $-\log_b t$

Question No: 25 (Marks: 3)

Differentiate:

$$f(t) = (t^3 + 4)^4$$
$$f'(t) = 4(t^3 + 4)^3 \cdot \frac{d}{dx}(t^3 + 4)$$
$$f'(t) = 4(t^3 + 4)^3 \cdot 3t^2$$
$$f'(t) = 12t^2(t^3 + 4)^3$$

Question No: 26 (Marks: 5)

$$\sqrt{13x^2 - 5x + 8}$$

Differentiate

$$f'(x) = \frac{1}{2\sqrt{13x^2 - 5x + 8}} \frac{d}{dx} 13x^2 - 5x + 8$$
$$f'(x) = \frac{1}{2\sqrt{13x^2 - 5x + 8}} \cdot 26x - 5$$

Question No: 27 (Marks: 10)

Differentiate the following function

$$f(x) = x^3 \cdot e^{\frac{1}{x}}$$

$$f'(x) = 3x^2 \cdot e^{\frac{1}{x}} + x^3 \cdot e^{\frac{1}{x}} - \frac{1}{x^2}$$

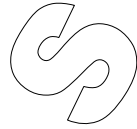
$$f'(x) = 3x^2 \cdot e^{\frac{1}{x}} - \frac{x^3 \cdot e^{\frac{1}{x}}}{x^2}$$

$$f'(x) = e^{\frac{1}{x}} \left[3x^2 - \frac{x^3}{x^2} \right]$$

$$f'(x) = e^{\frac{1}{x}} \left[\frac{3x^4 - x^3}{x^2} \right]$$

$$f'(x) = xe^{\frac{1}{x}} \left[\frac{3x^3}{x^2} - \frac{x^2}{x^2} \right]$$

$$f'(x) = xe^{\frac{1}{x}} [3x - 1] \text{Ans}$$



add